

$\text{\LaTeX}$ The Latency of Style: Assessing Stylistic Agency in Real-Time Diffusion for Live Audio-Visual Performance

K. Wolf¹

¹Film University Babelsberg KONRAD WOLF, Germany

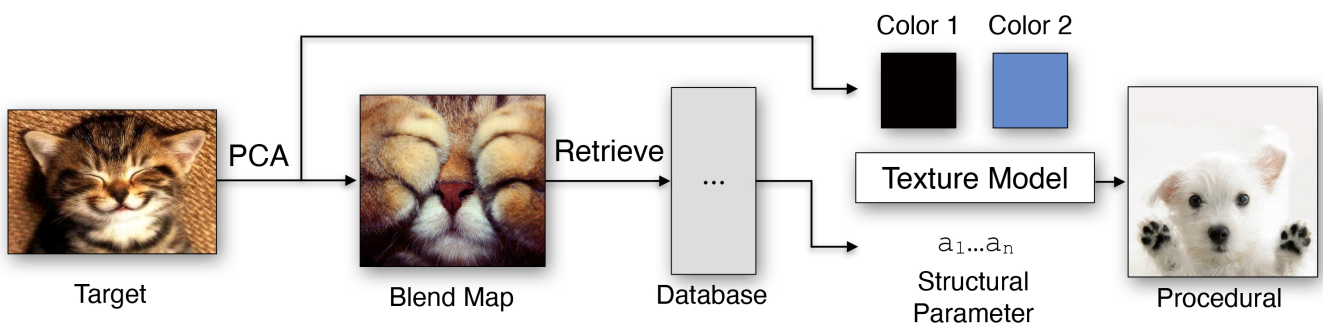


Figure 1: Teaser Image. Explain the image.

Abstract

The abstract should be between 100-150 words and should be a succinct and stand alone description. An abstract is neither a summary nor an outline of the paper. For example, you can follow these four sentences [JBB*93]: State the problem, Why is it a problem and why is it interesting? Your solution and achievements. What follows from your solution?

1. Introduction

Recent advances in diffusion models expanded the opportunities and use cases for AI-generated visual content. While early diffusion systems needed seconds or even minutes to create an image, current models are close to real-time generation. This advantage is especially interesting for artists in the field of audio-visual performances, where they slowly start to integrate these systems into their workflow.

TouchDesigner is a node-based visual scripting environment that is widely used for interactive live installations and performances. It allows for the integration of other software and data from hardware components and facilitates real-time rendering and audio-reactivity. The integration of diffusion models into this workflow additionally enables artists to blend their art with AI-generated content in real-time, making visuals more versatile. This is possible through the open-source pipeline called StreamDiffusion. Instead of changing the actual AI models, StreamDiffusion changes how the data flows, using pipeline-level tactics to improve latency, fps, and visual stability.

Despite those new developments, it remains difficult to gain full control over the generated output. Existing workflows primarily rely on standard text prompts and predefined model behavior which artists modulate externally using audio and video data. While this offers dynamic interactivity, it remains unclear to what extent this pipeline allows artists to include their individual handwriting. Furthermore, because StreamDiffusion relies heavily on baked TensorRT acceleration engines for real-time speed, integrating and swapping Low-Rank Adaptations (LoRAs) fluidly is highly limited, which further limits the visual guiding of the model.

For this reason the following research question is addressed: To what extent can artists leverage StreamDiffusion's parameters and custom LoRAs in TouchDesigner to assert stylistic agency without compromising real-time performance metrics?

To answer this question, a Streamdiffusion pipeline is implemented into TouchDesigner to identify and evaluate existing control methods like text prompting and audio/video input modulation. A further attempt is to incorporate a custom-trained Low-Rank Adaptation (LoRA). These control methods are then evalu-

ated through qualitative observations on artistic control and also quantitatively, through evaluation metrics, measuring their effect on performance parameters like latency and fps.

2. Related Work

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris at congue arcu. Nunc sollicitudin fringilla nibh, congue interdum justo. Nunc non lacus nisi. Curabitur lacus mi, posuere vitae porttitor quis, mollis eget magna. Nam egestas lorem maximus justo tempor, non tristique ipsum dapibus. Fusce vel ultricies arcu. Morbi in quam in mauris efficitur vulputate. Quisque eu diam quis orci sagittis convallis. Pellentesque a augue sollicitudin, pellentesque nisi ut, vestibulum tortor. Nunc vitae dapibus sapien. Suspendisse a nisi ut enim cursus viverra eget in ipsum. Nullam pellentesque sollicitudin dolor, et pretium dolor egestas vel. Etiam dapibus lobortis semper. Ut vitae porttitor dui, id hendrerit nulla. Ut euismod faucibus turpis, et interdum felis sagittis id. Donec at arcu eget est elementum facilisis.

3. Main Content - Name and Structure the Main Content According to your Paper

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris at congue arcu. Nunc sollicitudin fringilla nibh, congue interdum justo. Nunc non lacus nisi. Curabitur lacus mi, posuere vitae porttitor quis, mollis eget magna. Nam egestas lorem maximus justo tempor, non tristique ipsum dapibus. Fusce vel ultricies arcu. Morbi in quam in mauris efficitur vulputate. Quisque eu diam quis orci sagittis convallis. Pellentesque a augue sollicitudin, pellentesque nisi ut, vestibulum tortor. Nunc vitae dapibus sapien. Suspendisse a nisi ut enim cursus viverra eget in ipsum. Nullam pellentesque sollicitudin dolor, et pretium dolor egestas vel. Etiam dapibus lobortis semper. Ut vitae porttitor dui, id hendrerit nulla. Ut euismod faucibus turpis, et interdum felis sagittis id. Donec at arcu eget est elementum facilisis.

3.1. Method

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris at congue arcu. Nunc sollicitudin fringilla nibh, congue interdum justo. Nunc non lacus nisi. Curabitur lacus mi, posuere vitae porttitor quis, mollis eget magna. Nam egestas lorem maximus justo tempor, non tristique ipsum dapibus. Fusce vel ultricies arcu. Morbi in quam in mauris efficitur vulputate. Quisque eu diam quis orci sagittis convallis. Pellentesque a augue sollicitudin, pellentesque nisi ut, vestibulum tortor. Nunc vitae dapibus sapien. Suspendisse a nisi ut enim cursus viverra eget in ipsum. Nullam pellentesque sollicitudin dolor, et pretium dolor egestas vel. Etiam dapibus lobortis semper. Ut vitae porttitor dui, id hendrerit nulla. Ut euismod faucibus turpis, et interdum felis sagittis id. Donec at arcu eget est elementum facilisis.

3.2. Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris at congue arcu. Nunc sollicitudin fringilla nibh, congue interdum

justo. Nunc non lacus nisi. Curabitur lacus mi, posuere vitae porttitor quis, mollis eget magna. Nam egestas lorem maximus justo tempor, non tristique ipsum dapibus. Fusce vel ultricies arcu. Morbi in quam in mauris efficitur vulputate. Quisque eu diam quis orci sagittis convallis. Pellentesque a augue sollicitudin, pellentesque nisi ut, vestibulum tortor. Nunc vitae dapibus sapien. Suspendisse a nisi ut enim cursus viverra eget in ipsum. Nullam pellentesque sollicitudin dolor, et pretium dolor egestas vel. Etiam dapibus lobortis semper. Ut vitae porttitor dui, id hendrerit nulla. Ut euismod faucibus turpis, et interdum felis sagittis id. Donec at arcu eget est elementum facilisis.

3.3. Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris at congue arcu. Nunc sollicitudin fringilla nibh, congue interdum justo. Nunc non lacus nisi. Curabitur lacus mi, posuere vitae porttitor quis, mollis eget magna. Nam egestas lorem maximus justo tempor, non tristique ipsum dapibus. Fusce vel ultricies arcu. Morbi in quam in mauris efficitur vulputate. Quisque eu diam quis orci sagittis convallis. Pellentesque a augue sollicitudin, pellentesque nisi ut, vestibulum tortor. Nunc vitae dapibus sapien. Suspendisse a nisi ut enim cursus viverra eget in ipsum. Nullam pellentesque sollicitudin dolor, et pretium dolor egestas vel. Etiam dapibus lobortis semper. Ut vitae porttitor dui, id hendrerit nulla. Ut euismod faucibus turpis, et interdum felis sagittis id. Donec at arcu eget est elementum facilisis.

4. Formatting your paper

4.1. Bullet Points

The following is an example for a bullet points list. The contributions of this paper are

- my greate archivement number one,
- a second awesome insight,
- and last but not least the third contribution.

4.2. Figures

All graphics should be centered.

If your paper includes images, it is important that they are of sufficient resolution to be faithfully reproduced.

Instructions from the original conference template (for the ACS FUB it is not that relevant):

To determine the optimum size (width and height) of an image, measure the image's size as it appears in your document (in millimeters), and then multiply those two values by 12. The resulting values are the optimum x and y resolution, in pixels, of the image. Image quality will suffer if these guidelines are not followed.

Example 1: An image measures 50 mm by 75 mm when placed in a document. This image should have a resolution of no less than 600 pixels by 900 pixels in order to be reproduced faithfully.

This is an exemplary reference to Figure 3, which shows the incredible test image from the Eurographics conference.



Figure 2: Here is a sample figure. All figures must have sensible captions and must be referenced in the text.

4.3. Tables

A simple table might look like the following. However there are several ways to create and layout tables - it is up to you to style a table up to your liking. This is a reference to Table 1.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: Here is a sample table. All tables must have sensible captions and must be referenced in the text.

4.4. Formulas

In L^AT_EX it is fairly easy to add mathematical symbols and formulas. See the following as example [wik19]:

$$a = bq + r \quad (1)$$

which is true if a and b are integers with $b \neq c$.

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \quad (2)$$

4.5. Footnotes

Please do not use footnotes at all.

4.6. References

References are layouted automatically. Add the needed entry to the bibtex file and use the citation command to reference the entry.

If you don't want to mention the author's name just add the citation to the end of the sentence [Per85]. You can also cite more than one references at the same time [GSH18, Per85, SLKD15].

Sometimes you want to refer to the authors name. With this template you need to write out the names manually. For more than two authors you only use the name of the first author and abbreviate the remaining names with "et al."

An example for more than two authors:

Raul et al. [RRDR09] thoroughly review the different approaches for segmentation algorithms. Our problem requires the segmentation of color images, summarized by Cheng et al. [CJSW01].

An example for two authors:

Also aiming for a fragmented aesthetic appeal by transforming original image data "piecewise", Collomosse and Hall [CH03] render from salient image features a cubist version of an input. Similarly, Lai and Rosin [LR13] loosely aim for a cubist appearance by segmenting an input and by simplifying the segments into constant colors, while maintaining global structures.

An example for one author:

Artistically, we are inspired by the art of Wehrli [Weh03], who manually creates image decompositions and rearrangements.

5. Conclusion

Please direct any questions to the production editor in charge of these proceedings.

References

- [CH03] COLLOMOSSE J. P., HALL P. M.: Cubist style rendering from photographs. *IEEE Transactions on Visualization and Computer Graphics* 9, 4 (2003), 443–453. 3
- [CJSW01] CHENG H., JIANG X., SUN Y., WANG J.: Color image segmentation: Advances and prospects. *Pattern Recognition* 34, 12 (2001), 2259–2281. 3
- [GSH18] GEBHARDT C., STEVŠIĆ S., HILLIGES O.: Optimizing for aesthetically pleasing quadrotor camera motion. *ACM Transactions on Graphics* 37, 4 (2018), 90:1–90:11. 3
- [JBB*93] JOHNSON R. E., BECK K., BOOCH G., COOK W., GABRIEL R., WIRFS-BROCK R.: How to get a paper accepted at oopsla (panel). *SIGPLAN Notices* 28, 10 (1993), 429–436. 1



Figure 3: Two-column width figure

- [LR13] LAI Y.-K., ROSIN P. L.: Artistic rendering enhancing global structure. *The Visual Computer* 30, 10 (2013), 1179–1193. [3](#)
- [Per85] PERLIN K.: An image synthesizer. *SIGGRAPH Computer Graphics* 19, 3 (1985), 287–296. [3](#)
- [RRDR09] RAUT S., RAGHUVANSHI M., DHARASKAR R., RAUT A.: Image segmentation — a state-of-art survey for prediction. In *Proceedings of the International Conference on Advanced Computer Control* (2009), IEEE, pp. 420–424. [3](#)
- [SLKD15] SEMMO A., LIMBERGER D., KYPRIANIDIS J. E., DÖLLNER J.: Image stylization by oil paint filtering using color palettes. In *Proceedings of the Workshop on Computational Aesthetics* (2015), Eurographics Association, pp. 149–158. [3](#)
- [Weh03] WEHRLI U.: *Tidying up art*. Prestel, 2003. [3](#)
- [wik19] Latex/advanced mathematics. http://www.ananth.in/Notes_files/lmtut.pdf, 2019. Viewed June 26, 2026. [3](#)